Tyler Busching

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EDUCATION

SUNY Farmingdale State College **Farmingdale, NY**
Bachelor of Science in Computer Engineering Technology *Expected May 2027*
- Cumulative GPA: 3.67/4.0, Dean's List
- Relevant Coursework: Electric Circuits, Network Analysis, Digital Electronics, C++, VHDL, Assembly, Microprocessors, Semiconductor Devices, FPGA Design

Suffolk County Community College **Selden, NY**
Associate in Arts with Distinction *Aug 2022 – May 2024*
- Cumulative GPA: 3.6/4.0, Dean's List

TECHNICAL SKILLS

Languages: C++, VHDL, Assembly
Hardware: Arduino, FPGA (Vivado), Microcontrollers, Embedded Systems, Oscilloscopes, Multimeters, Logic Analyzers
Software: NI Multisim, MPLAB IDE, Proteus, Arduino IDE, Visual Studio, UNIX OS, Office 365, Adobe Suite
Other: PC Building & Repair, Process Mapping, BIOS Configuration, System Troubleshooting, Technical Design

TECHNICAL EXPERIENCE

IEEE Student Branch – SUNY Farmingdale State College **Farmingdale, NY**
Student Member *2026 – Present*
- Soldered and analyzed photoresistor-based circuits using voltage divider principles
- Assembled and debugged a handheld RF communication circuit; currently collaborating on an ultrasonic levitation system using 40kHz transducers (signal amplification, waveform control)
- Designed and implemented an Arduino-based temperature/humidity monitoring system (DHT11) with real-time output and calibration

PROJECTS

Custom PC Building & System Optimization *2018 – Present*
- Built and configured 20+ custom desktop systems, including BIOS tuning, CPU/GPU overclocking, thermal optimization, cable management, and peripheral integration
- Diagnosed hardware/software issues (boot failures, driver conflicts, overheating) and selected components based on performance, compatibility, and budget constraints

24-Hour Digital Clock Design *Fall 2025*
- Designed a digital display clock using counters, decoders, and flip-flops, tested and validated using NI Multisim

Arduino-Based Embedded Sensor Systems *Winter 2025*
- Engineered Arduino-based circuits for real-time data display, implementing analog signal reading and calibration

C++ Programming *Spring 2025*
- Wrote C++ programs to simulate electrical circuits using network analysis principles
- Developing a terminal-based financial tracking system to analyze user spending patterns (weekly/monthly)

WORK EXPERIENCE

Bourne Mansion, Timber Point Country Club **Great River, NY**
Catering Captain, Assistant Captain, Server *April 2023 – Present*
- Led coordination of 250+ guest events, managing teams of 10+ under strict time constraints
- Maintained efficiency in high-pressure environments, ensuring smooth execution of operations

ADDITIONAL EXPERIENCE

Freelance Multimedia & Creative Design *2019 – Present*
- Created 50+ digital and physical assets (graphics, videos, custom artwork) using Adobe Photoshop, After Effects, and studio art equipment, generating $2,000+ in revenue, managing projects independently
- Awarded NYS Certificate of Merit: 29th NYSATA Legislative Student Art Exhibit